

Resolution Geometry

The Groupoid of Resolution:

From the Kernel Pair to Global Transport, Bottom-Up and Top-Down

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Abstract

Resolution Geometry (RG) is the study of one groupoid and its representations. Top-down, the equality-of-law relation of a joint observation law is a groupoid, the kernel groupoid $\mathcal{K}$. Its orbits are the observational fibres, its quotient is the observable space, and its tangent is the latent kernel. Bottom-up, three kinds of structure are declared: groupoids of known indistinguishabilities, groups of physical operations, and second-order latent data. The two directions meet at the kernel. The stabilizer of the law modulo its gauge part gives the effective symmetry Γ ; natural modules descend and are Γ -invariant; declared groupoids are complete exactly when they exhaust the fibres.

This paper states the whole theory as the ladder climbed by this one groupoid:

- the exact layer and the symmetry layer;
- the 1-jet isotropy, which is the mature form of the early “automorphisms of resolution”;
- natural tensors and the observable transport groupoid;
- strata and non-invertible specialization;
- latent transport;
- isotropy forcing;
- closure.

It also shows how each declared hypothesis (H-reg, H-unif, H-hell, H-task, H-cov, H- Γ) acts on the groupoid. Three earlier branches are cut by explicit counterexamples:

1. The resolved–null coupling vanishes identically under the whitening (Čencov) metric. It measures the position of a declared protocol metric relative to the noise metric. It is not a law invariant.
2. The “principal $\text{Aut}(G)$ -bundle” over a varying operating point exists only on level sets of continuously varying invariants. The law-natural structure group is the constant $O(r) \times O(q)$.
3. Local automorphisms of (R, N, Σ) are isotropies of the 1-jet and need not integrate to symmetries of the law.

The selected natural metric changes the observable holonomy: on the same qubit slice the SLD metric has curvature $+1$ and the BKM metric is negatively curved. The declared covariance changes the latent holonomy. An eight-check battery passes 8/8 on five seeds, and two canaries are caught.

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1 One groupoid, two directions

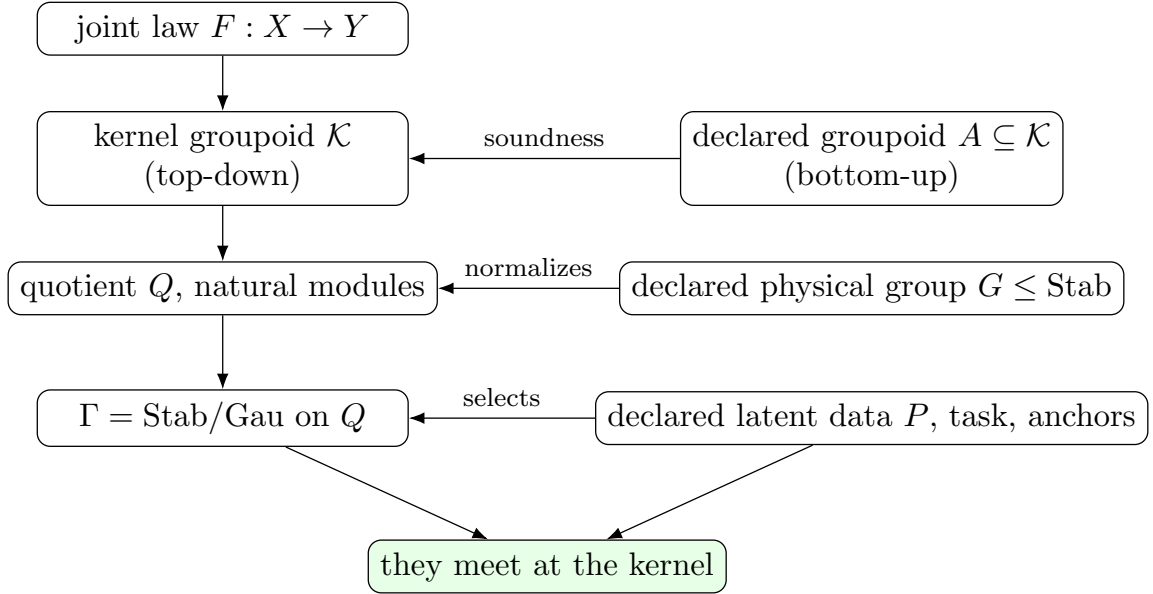


Figure 1: Top-down structure is computed from the law; bottom-up structure is declared. RG is the account of where the two agree and what each hypothesis adds.

An observation system is $(X, \{F_i\}, \Psi)$: a latent space, protocols with values in law spaces, and a declared group of outcome relabelings. Its joint law is $F = (F_i)$. Everything below is a groupoid, a functor between groupoids, or a representation of a groupoid. Each layer is a rung group of the theorem ladder [1]. Proofs that appear in the ladder or its module papers are cited, not repeated; new statements are proved here and tested in the battery GR01–GR08.

2 Level 0: the kernel groupoid (top-down)

Definition 2.1. $\mathcal{K} = X \times_Y X \rightrightarrows X$, with arrows $x \rightarrow x'$ iff $F(x) = F(x')$. Its orbits are the observational fibres, and its orbit space $Q = X/\mathcal{K} \cong \text{im } F$ is the observable space.

Theorem 2.2 (L1–L4). $\mathcal{K}$ is a groupoid, and fibre-constant maps factor uniquely through Q . For a joint law, $\mathcal{K} = \bigcap_i \mathcal{K}_{F_i}$. A garbling M gives an inclusion $\mathcal{K}_F \hookrightarrow \mathcal{K}_{M \circ F}$, which defines a functor of groupoids.

Proposition 2.3 (Lie groupoid on regular strata; L12–L13). Under $H\text{-reg}$, $\mathcal{K}|_S$ is a regular Lie groupoid. Its orbits are the fibres of the submersion π , and its Lie algebroid has anchor image $V = \ker DF$, the latent kernel. Every natural module annihilates V and is $\mathcal{K}$ -basic.

This is the only “null” that is law-intrinsic: parameter directions the law cannot see.

3 Level 1: symmetry and declared groupoids

Theorem 3.1 (Exact sequence; L6–L8). $\text{Stab} = \{(\varphi, \psi) : F \circ \varphi = \psi \circ F\}$ normalizes $\mathcal{K}$, and the gauge group Gau (global bisections of $\mathcal{K}$) is normal in it. The effective symmetry $\Gamma = \text{Stab}/\text{Gau}$ acts faithfully on Q :

$$1 \rightarrow \text{Gau} \rightarrow \text{Stab} \rightarrow \Gamma \rightarrow 1,$$

and the sequence need not split. Relabelings of counted channels are trivial (count rigidity).

Theorem 3.2 (Declared groupoids; L9–L11). A declared groupoid A is sound iff $A \subseteq \mathcal{K}$; then $X/A \rightarrow Q$. It is complete iff this map is a bijection, that is, iff A is wide in each fibre. If a declared physical group G normalizes A , the completeness locus is G -invariant.

The meeting point. Bottom-up data are sound only when they land inside $\mathcal{K}$ (A) or inside Stab (G). Top-down data are complete only when the declared data exhaust them. RG statements are the ones that hold on both sides.

4 Level 2: the 1-jet and its isotropy

At $x \in S$ the law has a first-order jet. For a locally Gaussian law with mean response $J = D\mu_x$ and noise Σ , the jet is (J, Σ) .

Definition 4.1 (Jet isotropy groupoid). The arrows from the jet at x to the jet at x' are the pairs (a, L) with

$$a : T_x X \rightarrow T_{x'} X, \quad L \in GL(O), \quad LJ_x = J_{x'} a, \quad L\Sigma_x L^\top = \Sigma_{x'}.$$

The isotropy group at x is Iso_x .

Theorem 4.2 (Structure of jet isotropy). Write $r = \text{rank } J$, $q = \dim O - r$, $R = \text{im } J$, and work in whitened coordinates ($\Sigma = I$). Then L preserves R and its Euclidean complement and acts as $(A_R, B_N) \in O(r) \times O(q)$. The latent part a is fixed by A_R on $T_x X/V$ and is free on V , as $GL(V)$ together with shears into V . Hence

$$\text{Iso}_x \cong (O(r) \times O(q)) \times (GL(V) \ltimes \text{Hom}(T_x X/V, V)),$$

and the second factor is the infinitesimal gauge.

Proof. $L\Sigma L^\top = \Sigma$ with $\Sigma = I$ makes L orthogonal. $LJ = Ja$ forces $L(R) = R$, hence L preserves $R^\perp$. On the quotient $T_x X/V \cong R$ via J , a is conjugate to A_R . On V any a with $Ja|_V = 0$ is allowed. $\square$

Remark 4.3 (Mature form of the early automorphism axiom). The early “CTMT automorphisms of resolution” $\varphi^*(O; R, N, \Sigma) = (O; R, N, \Sigma)$ are the L -components of Iso_x : outcome relabelings of the 1-jet. The observation-side axiom is not primitive. It is the linearization at one point of the stabilizer of the law.

No-Go 4.4 (Jet isotropy does not integrate; GR05). For $\mu(t) = (t_1, t_2, k t_1 t_2)$ with $\Sigma = I$, every rotation of the whitened resolved sector lies in Iso_x . For $k = 0$ it integrates to an element of Stab : the rotated image is the image, with defect 0. For $k = 0.8$ it does not: the rotated image leaves the law family by 0.029 at second order. Global symmetry is Γ , computed from the law; it is not read off the jet.

4.1 The protocol-metric module and the coupling

The early papers fix a reference metric g on O , set $N = R^{\perp_g}$, and decompose Σ into blocks (Σ_R, Σ_N, C) .

Theorem 4.5 (The coupling is relative; GR01). *For $g = \Sigma^{-1}$ (whitening, the Čencov-canonical choice), $C \equiv 0$: in every g -orthonormal adapted frame $\Sigma = I$. For a general declared g , $C = 0$ iff $R^{\perp_g} = R^{\perp_{\Sigma^{-1}}}$. The coupling is the relative position of the protocol metric and the noise metric. It is a non-natural module, a property of the declared g .*

Proof. In a Σ^{-1} -orthonormal frame, Σ is the identity, so its off-diagonal block vanishes. For general g , the R - N block of Σ vanishes iff N is Σ^{-1} -orthogonal to R . $\square$

On 200 random systems: $\max |C| = 0$ under whitening, median $|C| = 6.3$ under $g = I$, and no violation of the characterization.

Theorem 4.6 (Coupling efficiency is data processing; GR02, L28). *The Σ^{-1} -orthogonal projection onto R is sufficient for the parameter, with no information loss. The g -orthogonal projection is a garbling; it loses information iff $C \neq 0$, and in the just-identified case the loss in log-volume equals the sector mutual information $-\frac{1}{2} \sum_i \log(1 - \rho_i^2)$ (identity error 0).*

The efficiency, mutual-information and classification results of the early papers are therefore correct statements about a declared protocol metric and its projection estimator. Their moduli count $r q + r + q$ classifies triples (g, R, Σ) . They are retained as the *protocol-metric module*.

Proposition 4.7 (Two nulls; GR03). *The latent kernel $V = \ker J$ is the radical of every natural metric and is unchanged when Σ is altered on the observation null (defect 0). The observation null $R^{\perp_g}$ moves with the declared metric (median shift 0.43). The kernel groupoid depends only on the law; the observation null depends on g .*

5 Level 3: natural tensors and observable transport

Theorem 5.1 (Natural descent; L13–L14, L22). *Natural modules (pullbacks from law space) descend to Q and are Γ -invariant. In particular the Fisher metric $\bar{g}$ is a tensor on Q , and so are its Levi-Civita connection and, on regular embedded strata, the induced connection $P\nabla$ (L56).*

Definition 5.2 (Observable transport groupoid). The observable transport groupoid has as objects the points of a regular stratum of Q , and as arrows the parallel transports of the natural connection along paths. Its isotropy at q is the holonomy group, and its infinitesimal generator is the curvature (Ambrose–Singer) [7].

Theorem 5.3 (Frames and structure group; GR04). *In whitened adapted frames every chart has type (r, q) with $C = 0$. The frame bundle therefore reduces globally on each regular stratum to $O(r) \times O(q)$, a G -structure that always exists. The invariants $\tau_g(x)$ of a declared protocol metric vary continuously with x (minimal step 1.24 along a path). The early “principal $\text{Aut}(G)$ -bundle” exists only on level sets of τ_g ; between distinct operating points no Aut -valued transition exists. Variation along the base is carried by the natural tensor and its connection, not by transition maps.*

Proposition 5.4 (Edge certification and Čech gluing; GR06, retained). • *A clean loop product does not certify its edges: $D^{-1}D = I$ with both edges inadmissible (residual 1.36, cancellation 0).*

- *Admissible edges can still carry the Čech defect $\Omega = -I$.*
- *Admissible holonomy stays bounded under iteration (norms 4.06, 4.29, 4.00), while inadmissible defects amplify ($4.3 \rightarrow 1.1 \times 10^{10}$).*

A coherent atlas is a Čech 1-cocycle valued in the structure group, and edges are certified one by one.

Theorem 5.5 (H-task changes observable holonomy; GR07). *For quantum protocols the natural metric is selected by the declared task (L31–L34). On the qubit slice $\rho(a, b) = \frac{1}{2}(I + a\sigma_z + b\sigma_x)$ the Gaussian curvature is:*

- *1.000 at three points for SLD (estimation);*
- *-0.025 , -0.089 and -0.126 at the same points for BKM (Stein testing).*

Small-loop holonomy is curvature times area, so the observable transport groupoid depends on the task. The groupoid $\mathcal{K}$ and the group Γ do not.

6 Level 4: strata and specialization

Theorem 6.1 (Stratified category; L17–L21). *Under H-unif:*

- *kernels are upper semicontinuous at walls;*
- *a specialization arrow $Q_\varepsilon \rightarrow Q_0$ exists iff $\mathcal{K}_\varepsilon \subseteq \mathcal{K}_0$, and is a limit correspondence otherwise;*
- *specializations compose;*
- *regular symmetries descend iff they normalize $\mathcal{K}_0$, and the wall group may be larger.*

The global object is the category whose arrows are the transport groupoids of the strata together with the non-invertible specialization arrows. Holonomy is defined within strata, and paths into a wall are compared after specialization.

Proposition 6.2 (What H-hell does). *H-hell does not change the groupoid. It makes the functor “natural module” continuous along specialization arrows: Hellinger distances, Fisher modules and risks specialize (L30). Without it, the specialization arrow exists but carries no distinguishability (L30, wall-tap WT05).*

Proposition 6.3 (Isotropy walls). *Where orbit type changes, as at the fold of the wall-tap rhythm channel, the quotient has a slice model. The first-order tangent is replaced by a one-sided tangent cone (L15–L16). A rank wall is certified from data only as an upper bound in declared precision units (L38, L89).*

7 Level 5: latent transport

Theorem 7.1 (No law-natural connection; L59). *Gau acts transitively on horizontal complements of V by vertical shears $\varphi = \text{id} + (B_2 - B_1)DF$. No connection is Gau-invariant when $V \neq 0$.*

Theorem 7.2 (What H-cov does; L60; GR08). *A declared covariance P selects $H_P = \text{im}(P DF^\top)$. It adds a new groupoid, the holonomy groupoid of the P -connection, whose isotropy is the vertical holonomy of closed observed loops. For the same law, $P = I$ gives holonomy 1.60×10^{-4} and $P = \text{diag}(1, 2, \frac{1}{2})$ gives 6.0×10^{-5} around the same loop. With a memory kernel the arrows are path-space (RTS) lifts. With the same P across a wall, the wall horizontal space lies in the limit of the regular ones.*

8 Level 6: isotropy forcing

Theorem 8.1 (H- Γ ; L48, L74–L75). *The isotropy Γ_q acts on $T_q Q$. If it contains $SO^+(1, n)$ with $n \geq 2$, the invariant cone is the light cone and the invariant forms are $\lambda\eta$. If relative Euclidean motions lie in Γ with irreducible isotropy, every natural metric is $\lambda^2 h$ with λ constant. Geometry is forced by representation theory of the isotropy of the effective symmetry.*

Proposition 8.2 (Anchors as groupoid rigidity; L70–L72). *A dilation lies in Stab unless a counted channel is present, because counts admit no relabeling. The metre is fixed exactly when the dilation leaves Γ . Multiplicative count loss puts it back.*

9 Level 7: closure as an orbit statement

Theorem 9.1 (Statuses as orbit types; L87–L90). *For a request, let $\mathcal{I}$ be the indistinguishability groupoid of the requested object under the declared protocol. The statuses read:*

- *LOCKED: the orbit is trivial (modulo declared isotropy) and certified;*
- *CORRESPONDENCE: the orbit is nontrivial, and the maximal correspondence is the orbit;*
- *VACUOUS: no object is requested by a nonzero task;*
- *STRUCTURAL WALL[ε]: the orbit type is unresolved at the window reached;*
- *ESCALATE: a finer window can decide the orbit type;*
- *INCONSISTENT: no orbit fits;*
- *OUTSIDE DOMAIN: no groupoid is declared.*

RG never chooses within a nontrivial orbit.

10 How each hypothesis acts on the groupoid

hypothesis	action
H-reg	$\mathcal{K} _S$ is a regular Lie groupoid; Lie algebroid anchor image V
H-sound, H-compl, H-coh	declared A lies in $\mathcal{K}$, exhausts orbits, and is normalized by G
H-nuis	fixes which part of $\mathcal{K}$ is treated as nuisance; the efficient metric is the Schur complement over it
H-unif	adds non-invertible specialization arrows between strata
H-hell	makes the natural-module functor continuous along specialization arrows
H-task	selects the natural metric, hence the observable connection and its holonomy
H-cov	adds the latent connection groupoid and its vertical holonomy
H- Γ	declares subgroups of Γ whose isotropy representation forces cone and metric
H-SI, H-med	remove dilations and medium fibres from Stab (rigid counted channels, anchored speed)
H-stat	places time translation in Γ on a source channel (transfer across outages)

11 Branches cut

1. *Observation-side morphism axiom as primitive.* Replaced by Stab of the law. The early axiom is the outcome part of jet isotropy (Theorem 4.2).
2. *Coupling as the primary invariant.* It vanishes under the canonical metric (Theorem 4.5). It is a protocol-metric module, and its efficiency theorem is data processing (Theorem 4.6).

3. $N = \ker J^\top$ as “the” null. The law-intrinsic null is the latent kernel V ; the observation null depends on the declared pairing.
4. *Principal* $\text{Aut}(G)$ -bundle over a varying base. It exists only on level sets of τ_g . The global structure is the $O(r) \times O(q)$ -reduction, together with natural tensors and their connection.
5. *Local automorphism = symmetry*. Jet isotropy need not integrate (GR05).
6. *Strict versus conformal “scale valve”*. Scale is decided by Γ : a dilation is in Stab unless a counted anchor removes it (L70–L72). It is not decided by a choice of covariance protocol.
7. *Sector-relative geometry*. Superseded by declared nuisance (H-nuis) and the efficient metric.

Retained without change:

- edge certification independent of monodromy;
- Čech gluing;
- amplification of inadmissible defects;
- the Schur structure of Fisher information;
- the incompleteness of canonical correlations (as protocol-metric invariants);
- Fisher insufficiency for covariance.

12 Place in the ladder

level	rungs
0 kernel groupoid	L0–L5, L12–L14
1 symmetry, declared groupoids	L6–L11
2 jet isotropy, protocol-metric module	L22–L28 (with Theorems 4.2–4.6 here)
3 natural tensors, observable transport	L13–L14, L22, L31–L34, L56
4 strata, specialization	L15–L21, L30, L38
5 latent transport	L57–L60
6 isotropy forcing, anchors	L48, L70–L75
7 closure	L87–L90

A Reproducibility

`rg_groupoid_attack.py` passes 8/8 on seeds 20261005, 1, 42, 777 and 2026. The canaries are a protocol metric set equal to the whitening metric, which leaves no coupling to find (GR01), and the same task twice, which leaves no holonomy gap (GR07). Both are caught. The other levels are tested in the module batteries [2, 3, 4, 5, 6].

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